Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Throughout this project, I expect to use loops, dataclasses, lists, and unit tests. Some programming skills I want to improve while completing this projects include, loops, and dataclasses.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the coding for the math portion of the project to be the easiest. It is simple math and can be accomplished in few lines of code. The hardest part will be the drafter portion of the project. I am not very familiar with drafter yet and will need to improve in that section along the way.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The drafter portion will require the most time and effort due to its complexity. I need to implement the initial code first in order to put the code into a drafter UI.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions that might interfere with my work may include social media and my social life. I could minimize these distractions by designating time to study in the library. Some resources I will need while working on the project include the internet, my computer, and dnd. I currently have no questions on the assignment requirements.

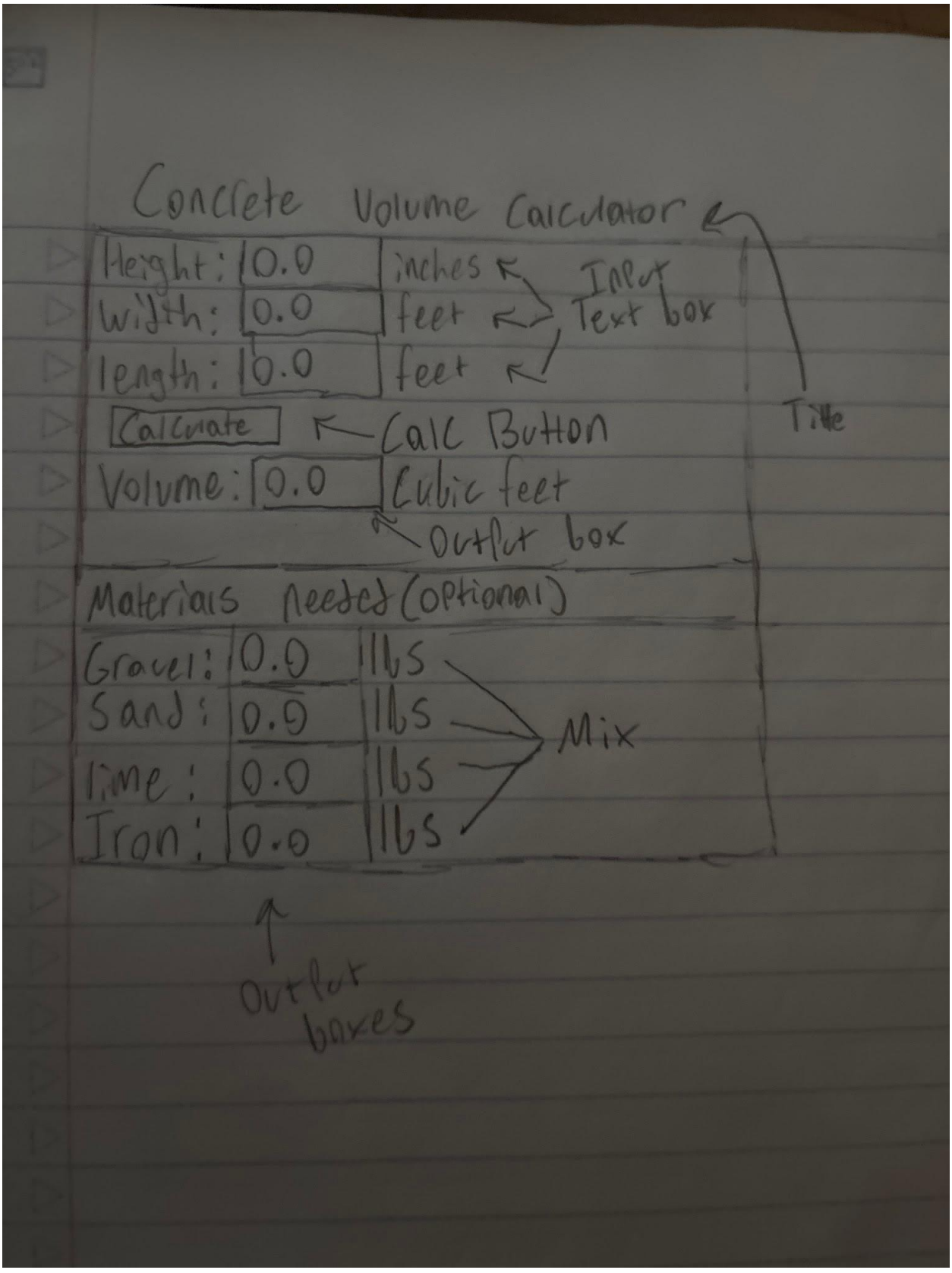
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Concrete Volume Calculator
Height: 0.0 inches
Width: 0.0 feet
length: 0.0 feet
Input Text box
Calculate
Calc Button
Title
Volume: 0.0 Cubic feet
Output box
Materials needed (optional)
Gravel: 0.0 lbs
Sand: 0.0 lbs
lime: 0.0 lbs
Iron: 0.0 lbs
Mix
Output boxes

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of my website is to take rough dimensions of a concrete pad and convert them to the total amount of concrete needed. This will help when ordering/purchasing concrete for the project.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	**Length: float** Stores the length of the pad **Width: float** Stores the width of the pad **Height: float** Stores the height/thickness of the pad **Current_volume: float** Stores calculated volume of the pad
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Length: float -> Measurement Width: float -> Measurement Height: float -> Measurement Volume: float -> Calculation of measurements
Routes (minimum 7) The names of your route functions	Index: Home page Calculator: calculator page Calculate_volume: calculates concrete needed Results: display concrete needed Delete_pad: remove saved pad

	Save_pad: stores pad Materials: calculated list of materials
Pages (minimum 5) A brief description of the pages in your site	Home: Intro to calculator Calculator page: actual calculator input page Results page: Output page Saves pads page: Stores projects Materials page: Calculated materials needed
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Textbox: Enter pad name Textbox: enter pad length Textbox: enter pad width Textbox: enter pad height/thickness Selectbox: select measurement type (feet/inches)
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Checks if measurements are greater than 0, if not it will display an error Checks if inches or feet are used, converts Checks if saved pad list is empty if it is it will display no saved pads
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	I could use a for loop to go through the saved pads list and go through the attributes of each and possibly calculate a total for all saved pads if needed.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top

of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM to assist me in this assignment.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.